

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I, Nethra P(192511012) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Nethra P

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications*.

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

- (a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.
- (b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

- (a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.
- (b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

- (a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.
- (b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?
- (c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

- (a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.
 - (b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.
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RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

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ASSESSMENT TOOL 2
Annexure A — Article Review

Article Citation Details

Authors	Yi Liu, Shuo Yang and Miaochao Chen
Title	Application of Decision Tree-Based Classification Algorithm on Content Marketing
Journal	Journal of Mathematics (Hindawi), Volume 2022, Article ID 6469054, pp. 1–10
Year of Publication	2022 (March)
DOI	https://doi.org/10.1155/2022/6469054
Database	Wiley Online Library / Hindawi (indexed, open access, CC-BY 4.0)
Domain / Application Area	Digital and content marketing analytics — classification of bank telemarketing campaign responses
Specific ML Problem Addressed	Binary classification of whether a bank customer will subscribe to a term-deposit product after a telemarketing campaign, using a decision-tree (C4.5) model benchmarked against five other classifiers

Task 1: Article Summary

(a) Full Citation, Domain and ML Problem

Liu, Y., Yang, S., & Chen, M. (2022). Application of Decision Tree-Based Classification Algorithm on Content Marketing. Journal of Mathematics, 2022, Article ID 6469054, 1–10. <https://doi.org/10.1155/2022/6469054>. The article belongs to the domain of digital/content marketing analytics, and specifically addresses a supervised binary classification problem: predicting, from a bank's telemarketing campaign data, whether a customer will subscribe to a term-deposit financial product. The dataset is a real bank content-marketing dataset originally collected by a Portuguese banking institution during outbound phone-call campaigns.

(b) Key Objective and Research Gap (150–200 words)

The authors observe that traditional, rule-of-thumb content-marketing strategies struggle to keep pace with the scale and complexity of modern customer data, often producing inaccurate targeting decisions. While machine learning has been widely adopted in other domains, the article identifies a gap in applying interpretable, human-reasoning-aligned models — specifically decision trees — to the content-marketing decision problem, where marketers need not just predictions but explainable rules they can act on. Most prior marketing-analytics work either used black-box models with little interpretability or applied decision trees without a systematic, multi-metric comparison against alternative classifiers. To fill this gap, the

authors propose using the C4.5 decision-tree algorithm — which can natively handle continuous attributes and produces human-readable if-then rules — on a real bank campaign dataset, and they benchmark it against five other machine-learning methods (k-nearest neighbours, linear SVM, neural network, Naïve Bayes, and random forest) using accuracy, precision, recall, and F1-score, so that both predictive strength and interpretability can be evaluated together.

Task 2: Methodology Analysis

(a) ML Technique(s) and Dataset

The core technique used is the C4.5 decision-tree algorithm, an extension of Quinlan's ID3 that selects splitting attributes using the Gain Ratio (and, for comparison, the Gini Index) rather than raw information gain, which reduces its bias toward attributes with many distinct values. C4.5 can also directly handle continuous-valued features by dynamically choosing threshold cut-points, and it applies Bayesian Error Pruning (BEP) — a top-down post-pruning method proposed by Quinlan — to trim branches that would otherwise cause the tree to overfit the training data, without requiring a separate pruning dataset. Five additional classifiers were used as benchmarks: k-nearest neighbours (k fixed at 3), a linear SVM (penalty parameter $C = 0.025$), a neural network (regularisation parameter fixed at 1, other hyperparameters at scikit-learn defaults), Naïve Bayes, and random forest (maximum depth fixed at 5).

The dataset is a bank campaign content-marketing dataset derived from a Portuguese banking institution's real telemarketing operations. It contains 50 user/campaign features (demographic details, prior contact history, and campaign-call attributes) collected to determine whether a contacted customer is likely to purchase (subscribe to) the offered product. For preparation, the authors performed feature correlation analysis (visualised as a heatmap) to understand redundancy among the 50 features, and they used a comparatively small training split — only 20% of the samples for training and the remaining 80% held out for testing — which is unusual and is itself worth scrutinising (see Task 3b). The tree's maximum depth was also tuned and its effect on accuracy studied using both the gain-ratio and Gini-index splitting criteria.

(b) Mapping to CO4 and Strength/Limitation

The article maps directly onto two CO4 topics simultaneously. First, it is a Decision Tree application, since the central model is C4.5, including its splitting-criteria (gain ratio/Gini), pruning (BEP), and depth-tuning mechanics. Second, it is an instance of Inductive Learning, because the tree is built purely by generalising if-then classification rules from labelled historical examples (past campaign outcomes) to predict unseen customers' behaviour — the hallmark of inductive, example-driven learning rather than deduction from prior domain axioms. The comparison against Naïve Bayes additionally brushes against Statistical Learning Methods, giving the study a broad footprint across CO4's syllabus themes.

One clear methodological strength is the multi-metric, multi-model benchmarking design: rather than reporting a single accuracy figure, the authors evaluate six different classifiers on four validation metrics (accuracy, precision, recall, F1-score), which gives a much more balanced picture of the decision tree's

relative strengths (e.g., where it wins on accuracy but is only second-best on F1). A significant limitation, however, is the unconventional 20%-train/80%-test split. Using only one-fifth of the data for training a decision tree — a model that is already prone to overfitting on small samples — raises real concerns about whether the reported performance would generalise to a genuinely held-out, larger real-world deployment, and the paper does not report cross-validation or confidence intervals to reassure the reader on this point.

Task 3: Results & Critical Evaluation

(a) Key Results and Comparative Table

The authors report that, across the four evaluation metrics (accuracy, precision, recall, and F1-score), the C4.5 decision tree achieved the strongest overall predictive performance among the six benchmarked models on the held-out test data. The neural network and random forest were the closest competitors, taking the runner-up position on F1-score and recall respectively; k-nearest neighbours was the second-best performer on F1-score; the linear SVM ranked second-best on both recall and F1-score; the neural network was the second-best performer on accuracy; and Naïve Bayes was the second-best performer on precision. A qualitative summary of the reported rankings is given below.

Model	Reported Rank / Notes
Decision Tree (C4.5)	Best overall — highest values across the four metrics (accuracy, precision, recall, F1)
Neural Network	Second-best on accuracy; strong showing overall
Random Forest (max depth = 5)	Second-best on recall; suboptimal on F1
k-Nearest Neighbours (k = 3)	Second-best on F1-score
Linear SVM (C = 0.025)	Second-best on recall and F1-score
Naïve Bayes	Second-best on precision

Note: the source article presents its precise numerical accuracy/precision/recall/F1 values in a dedicated results table (Table 1) and depth-vs-accuracy plots (Figure 5); the ranking summary above reflects the comparative conclusions the authors draw from that table rather than reproducing the table itself.

(b) Are the Metrics Realistic? Potential Biases

The reported dominance of the decision tree is plausible in isolation — C4.5 is well known to perform competitively on tabular, mixed-type marketing data — but several factors raise questions about how representative the numbers are. First, the 20%/80% train–test split is inverted relative to common practice (typically 70–80% is used for training), which increases the risk that the small training set happened to favour the decision tree's particular splitting behaviour on this one partition; the authors do not report repeated runs, k-fold cross-validation, or confidence intervals, so it is impossible to know whether the ranking would hold on a different random split. Second, the underlying bank-marketing data is known

(from similar public datasets of this kind) to be class-imbalanced — far fewer customers subscribe than decline — and the article does not explicitly discuss how class imbalance was handled, which can inflate accuracy while masking poor recall on the minority (subscribing) class that matters most to marketers. Third, hyperparameters for the comparison models (e.g., $k = 3$ for k -NN, random-forest depth capped at 5) appear to be fixed rather than tuned via grid search or cross-validation, which could understate the true competitiveness of those alternative models and make the decision tree's win look larger than it would be under a fairer, tuned comparison.

(c) Real-World Applicability and Scaling Challenges

A C4.5-style decision tree is well suited to real-world content-marketing deployment because its rules are transparent — a marketing team can read the tree's splits (e.g., "if prior-contact count $> X$ and account tenure $< Y$, then predict subscribe") and translate them directly into campaign-targeting policy, without needing a data-science intermediary to explain a black-box score. This makes it deployable for customer segmentation, next-best-offer targeting, churn prediction, and campaign-budget allocation in retail banking, telecom, and e-commerce settings. Scaling the approach to industry, however, faces several challenges: (i) data availability and quality — production systems must continuously refresh training data as customer behaviour and macroeconomic conditions drift, whereas the article uses a static historical snapshot; (ii) compute and maintenance — although a single tree is cheap to train, production pipelines need automated retraining, monitoring for concept drift, and version control, none of which the paper addresses; and (iii) ethical and regulatory issues — using customer demographic and financial-history features for marketing decisions raises fairness and privacy concerns (e.g., unintentionally encoding proxies for protected attributes such as age or income bracket), and financial-sector deployments would need to satisfy explainability and non-discrimination requirements under regulations such as GDPR or local banking-conduct rules.

Task 4: Reflection & Learning Outcome

(a) Three Key Insights Linked to CO4

- Insight 1 — Splitting criteria matter as much as the algorithm family: Comparing the gain-ratio and Gini-index variants of C4.5 within the same paper reinforced the CO4 concept that a decision tree's quality depends heavily on the impurity/split-selection measure chosen, not just on whether a tree is used at all — directly connecting to the syllabus topic of Decision Tree construction and attribute-selection measures.
- Insight 2 — Inductive learning is only as good as the representativeness of its training sample: The article's unusually small 20% training split showed me, concretely, how the Inductive Learning principle ("generalise a hypothesis consistent with the training examples") can break down in practice if the training sample is too small or unrepresentative, which links to the course's discussion of bias–variance trade-offs and overfitting in inductive learners.

- Insight 3 — Multi-metric evaluation is essential for imbalanced, real-world classification: Seeing the decision tree win on some metrics but lose on others (e.g., F1 vs. recall) to different competing models reinforced the CO4 statistical-learning idea that accuracy alone can be misleading, and that precision, recall, and F1-score together give a fuller picture of a classifier's suitability — especially important for the reinforcement-learning-adjacent theme of decision-making under uncertainty, where false negatives and false positives carry different real costs.

(b) Proposed Extension / Novel Experiment

If I were to extend this research, I would propose replacing the single, un-cross-validated 20/80 split with a stratified 5-fold cross-validation protocol combined with SMOTE-based oversampling of the minority (subscribing) class, and then replacing the plain C4.5 tree with an XGBoost gradient-boosted tree ensemble tuned via Bayesian hyperparameter optimisation. This is feasible because the dataset is small (tabular, 50 features) and the tools required (imbalanced-learn, XGBoost, scikit-optimize) are freely available, open-source, and computationally inexpensive to run even on a laptop, so the experiment would not require any new data collection — only re-analysis of the same or a similarly sourced bank-marketing dataset. The expected impact is twofold: (i) a more statistically reliable estimate of true generalisation performance (via cross-validation) that removes the single-split bias identified in Task 3b, and (ii) a meaningful lift in recall on the minority class (the customers most valuable to identify) without sacrificing the interpretability that content marketers rely on, since feature-importance rankings from a gradient-boosted tree remain human-interpretable in a way that, say, a deep neural network's weights are not.

Self-Assessment Against Rubric

Criteria	Weightage	How this review addresses it
Understanding of Article	20%	Task 1 restates the article's purpose, dataset, and research gap in original words
Summary	20%	Task 1(b) gives a concise, accurate 150–200 word summary of objective and gap
Critical Analysis	25%	Task 3(b)–(c) critiques the split ratio, possible class imbalance, and hyperparameter tuning
Use of Evidence	15%	Citation block, comparative table, and specific method/dataset details from the article support every claim
Organization	20%	Clear headings for Tasks 1–4 mirroring the assessment structure, tables for citation and results